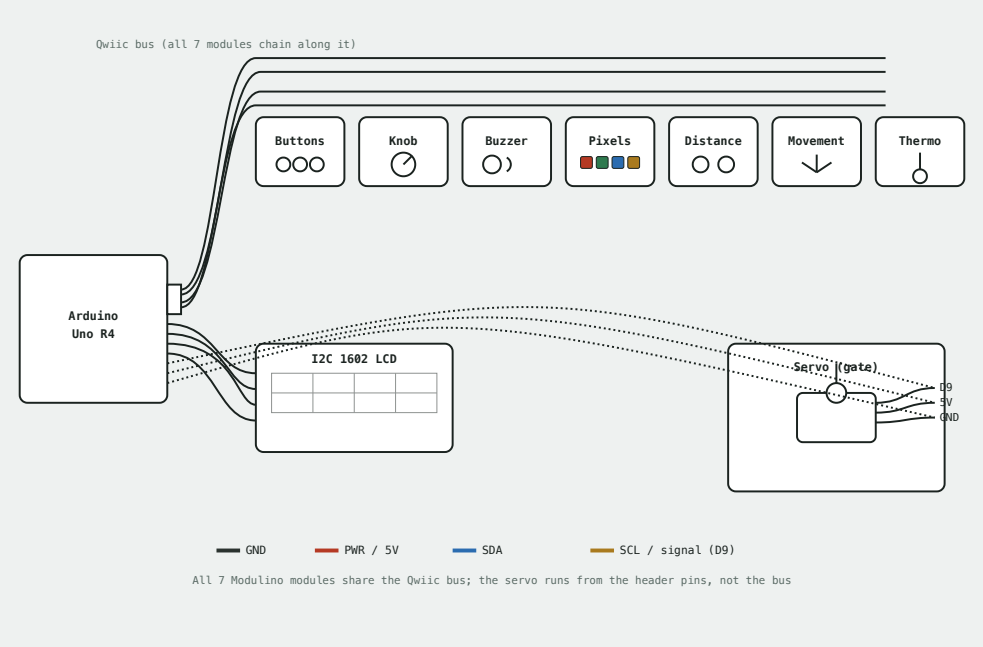
Mini weather & security station

Every module from projects 1–5, plus one new actuator: a servo “gate”. Button A cycles three screens — Weather, Parking, Gate — while the shake alarm from project 5 keeps watching in the background and can interrupt any of them.

Board · Arduino Uno R4 **Modules** · all 7 Modulino modules, I2C LCD, servo
New concepts · menu state machine, `Servo` library



All 7 Modulino modules share one Qwiic bus in any order. The LCD is wired exactly as in project 2. The servo is new: signal to pin 9, power to 5V, ground to GND (dashed lines, since they run from the Arduino’s header rather than the Qwiic bus).

PIN & CONNECTION REFERENCE

Component	Connects to
Buttons, Knob, Buzzer, Pixels, Distance, Movement, Thermo	Qwiic/I2C bus (any order)
I2C 1602 LCD	VCC→5V, GND→GND, SDA→SDA, SCL→SCL
Servo	Signal→D9, power→5V, ground→GND

Gotcha: a small hobby servo can usually run briefly from the Arduino’s own 5V pin, but if it twitches or resets the board under load, power it from a separate 5V supply and tie its ground to the Arduino’s GND.

REQUIRED LIBRARIES

- `Arduino_Modulino` — pulls in the sensor-vendor dependencies automatically
- `LiquidCrystal_I2C` — Library Manager
- `Servo` — bundled with the Arduino IDE

CODE REFERENCE

Screens as a state machine

Same `enum` idea as project 5, applied to navigation.

```
// --- Variables (top of sketch) ---
enum Screen { WEATHER, PARKING, GATE,
              SCREEN_COUNT };
Screen currentScreen = WEATHER;

// --- Inside loop() ---
if (changed && buttons.isPressed('A')) {
  currentScreen = (Screen)
    ((currentScreen + 1) % SCREEN_COUNT);
}
```

Read the buttons once per loop

`update()` only reports a change once, so share the result.

```
// --- Inside loop() ---
bool changed = buttons.update();
// ...inside the active screen:
if (changed && buttons.isPressed('C'))
  gateServo.write(gateAngle);
```

Gate — knob steps, servo moves

The knob counts finely, so react to direction (5° per click).

```
// --- Variables (top of sketch) ---
int gateAngle = 0;
const int GATE_STEP_DEG = 5;

// --- Inside loop() ---
int dir = knob.getDirection();
if (dir != 0)
  gateAngle = constrain(
    gateAngle + dir * GATE_STEP_DEG, 0,
180);
```

Alarm takes priority

While the alarm screen shows, the normal screens don’t run.

```
// --- Inside loop() ---
updateAlarm();
if (showingAlarmScreen) {
  // any button dismisses it
  return;
}
```

TRY IT YOURSELF

- Add a fourth screen that shows the alarm’s current state (Armed/Alarm) even when it hasn’t tripped.
- Make button B move the servo to a fixed “closed” position (0°) as a shortcut.
- Design your own 7th project that mixes modules in a way none of the six here did.